

Resource efficiency of nnU-Net for malignant liver mass segmentation on B-mode ultrasound

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Abstract

What is the minimum practical investment in training data and effort needed for deep learning AI to detect cancerous liver masses in ultrasound images? We address this question by training nnU-Net on the Annotated Ultrasound Liver (AUL) dataset at eight nested training set sizes (from 25 to 625 images) and by evaluating each subsequent model against a fixed 110-image test set. Mass segmentation improves steadily through 625 images (0.65 Dice) with no plateau. Per-class analysis, however, reveals that malignant mass detection reaches 0.77 Dice at 625 images, while benign mass detection achieves only 0.40 Dice. The detection rate of malignant masses comes to 97%, compared to just 70% for benign masses. Ablation experiments indicate that model capacity and training duration are not limiting factors at this dataset size. Increasing the training set size from 625 to just 1,500 images would in theory bring the Dice for malignant masses to better than 0.90, according to log curve estimates, while benign masses show no comparable path to useful performance through data scaling alone.

1 Introduction

Automated liver mass detection on B-mode ultrasound could expand screening and monitoring capabilities in point-of-care settings with limited access to specialists and expensive specialized infrastructure. Ultrasound data remains under-utilized in machine learning research relative to other imaging types (Al-sharid et al., 2025). Meanwhile, deep learning models typically require significant training investment, both in compute and human capital. This raises a practical question: how few compute resources and training images are needed to train a useful malignant liver mass detection model?

We address this through a resource efficiency analysis using nnU-Net (Isensee et al., 2021), whose self-configuring design reduces the burden not just of GPU compute but of the expert human labor typically needed to configure a model for given training goals. We train on the Annotated Ultrasound Liver (AUL) dataset at eight nested subset sizes (25 to 625 images), evaluating each model against a fixed test set.

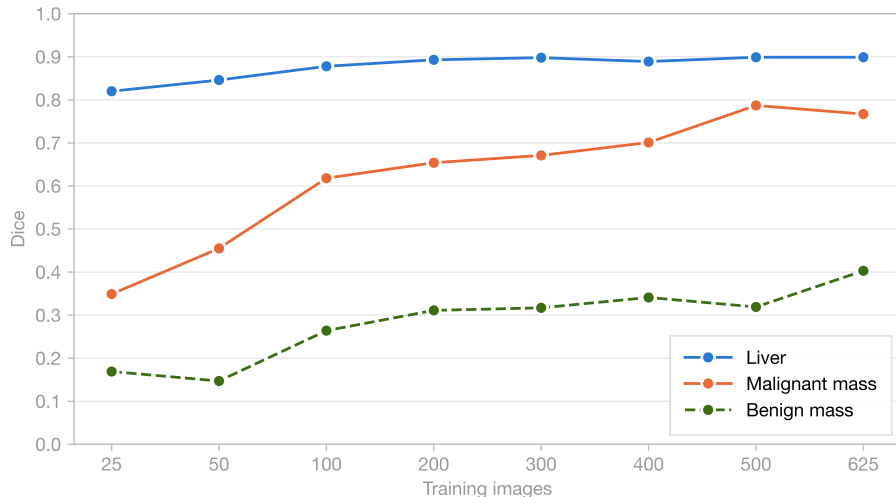


Figure 1: Dice coefficient by training set size and pathology type. Liver segmentation (blue) plateaus at 0.89 by 200 images. Malignant mass Dice (orange) climbs steadily to 0.77 at 625 images. Benign mass Dice (gray, dashed) remains below 0.41 across all sizes.

The aggregate results mask a clinically important distinction: malignant and benign masses behave very differently as a segmentation target. Per-class analysis reveals that malignant mass detection reaches 0.77 Dice with 625 training images and a 97% detection rate, while benign mass detection achieves only 0.40 Dice and 70% detection. We investigate the sources of this gap through analysis of lesion size, intensity contrast, and size-class interactions.

2 Related work

2.1 Liver mass segmentation studies

Liver tumor segmentation has been studied extensively on CT imagery; less so on MRI and ultrasound.

The canonical CT benchmark for liver tumor segmentation is the Liver Tumor Segmentation Benchmark (LiTS), which provides 131 annotated CT volumes (Bilic et al., 2023). On LiTS, nnU-Net achieves tumor Dice scores in the range of 0.78–0.82, and remains competitive with or superior to more complex architectures, including transformer-based models such as SwinUNETR and U-Mamba (Manghi et al., 2025).

On MRI, Hänsch et al. used an anisotropic 3D U-Net on late-phase DCE-MRI for HCC segmentation. They first trained 16 copies of the same model with different random seeds, then they selected the best performer, which had

mean Dice of 0.74 (Hänsch et al., 2022); the inter-rater agreement context was 0.78.

Wu et al. developed BACANet for liver boundary segmentation on ultrasound, evaluating on both a public dataset (AUL) and a larger, private dataset from Taizhou People’s Hospital. BACANet performs liver boundary segmentation only, not mass segmentation, achieving liver Dice of 0.92. (Wu et al., 2024).

Tupper and Gagné (Tupper and Gagné, 2025) trained U-Net on that same AUL data set, achieving a mass Dice of 0.51 without augmentation. This result reflects on a smaller training split and normal cases where mass Dice is trivially zero, so that our nnU-Net baseline of 0.65 on mass-present cases isn’t one-to-one comparable.

Natali et al. applied nnU-Net to hepatic tumor segmentation on intraoperative B-mode ultrasound in a surgical navigation context, achieving Dice of 0.84 (Natali et al., 2025). Our work addresses a different clinical setting: diagnostic B-mode imaging for screening and monitoring, where image quality, acquisition conditions, and clinical objectives differ.

2.2 nnU-Net on ultrasound

nnU-Net has become a de facto baseline for medical image segmentation across modalities, though its application to ultrasound remains limited. On automated breast ultrasound (ABUS), nnU-Net achieved a Dice of 0.81 with post-processing on a 100-case dataset (Tan et al., 2024).

More broadly, Alsharid et al. catalog 72 public ultrasound datasets and note that ultrasound remains under-utilized relative to CT and MRI in machine learning research (Alsharid et al., 2025). The nnU-Net framework has also been used as a component in federated learning experiments on ultrasound data (Deng et al., 2025).

2.3 Data efficiency in medical image segmentation

The question of how much training data is needed for a segmentation model has been addressed in several contexts. In 2016, J. Cho et al. applied learning curve analysis to medical image classification, varying training set sizes from 200 to as few as 5 CT images (Cho et al., 2016).

More recent studies have applied similar methods to segmentation. M. Cho et al. trained nnU-Net on progressively smaller subsets for pelvic multi-organ segmentation on MRI, finding stable performance down to 12 training images before a sharp drop (Cho et al., 2024).

These studies generally focus on high-contrast anatomical structures or use CT/MRI data. Our work extends this progressive subset approach to a harder target (liver masses on B-mode ultrasound) and provides per-class data efficiency curves that reveal markedly different data requirements by pathology type.

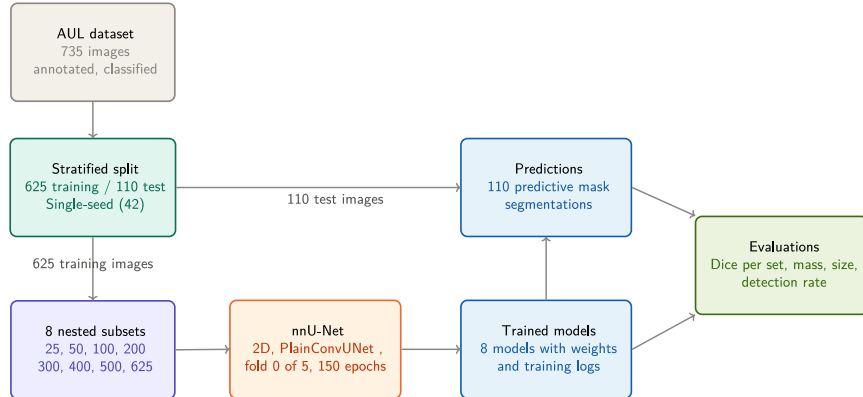


Figure 2: Study pipeline. The AUL dataset is split into fixed test and training sets, with nested subsets drawn from the training set. Each subset trains an independent nnU-Net model evaluated against the same test set.

In summary, while nnU-Net has been applied to liver tumors on CT and to other organs on ultrasound, our search identified no prior application to liver mass segmentation on diagnostic B-mode ultrasound, nor any resource efficiency analysis for this task.

3 Data and methods

We selected nnU-Net for this study because its self-configuring design eliminates manual architecture search and hyperparameter tuning, keeping compute costs to minimal training runs per configuration (Isensee et al., 2021). Isensee et al. subsequently demonstrated that many claims of architectures outperforming nnU-Net do not survive rigorous validation (Isensee et al., 2024). These properties make nnU-Net particularly suited to resource-constrained settings where extensive experimentation is impractical, and its strong baseline performance ensures that observed limitations reflect the difficulty of the task rather than suboptimal model configuration.

We selected the AUL dataset because it is, based on our search methodology¹, the largest publicly available liver ultrasound dataset with both liver and mass segmentation annotations (Xu et al., 2022).

¹Search methodology, training and analysis code, results, and end-to-end reproduction instructions are available at <https://github.com/roprice/liver-us-nnnet-baselines>.

3.1 AUL dataset

The Annotated Ultrasound Liver (AUL) dataset (Xu et al., 2022), contains 735 B-mode liver ultrasound images across three categories: 500 malignant masses, 150 benign masses, and 85 normal (no mass). Each image has expert-annotated polygon segmentations for liver and mass regions, along with outline polygons delineating the ultrasound scan sector boundary.

The dataset was split into 625 training images and 110 test images using a stratified 85/15 split (seed 42), preserving the category proportions across splits. The test set contains 65 malignant, 30 benign, and 15 normal cases.

The AUL dataset receives a Bronze rating on the SonoDQS quality scale (Alsharid et al., 2025), reflecting limited documentation of acquisition parameters (including device and transducer type) and annotation methodology rather than data quality per se.

3.2 Outline masking and preprocessing

The AUL dataset provides outline polygons marking the ultrasound scan sector boundary. Pixels outside the scan sector (black padding, UI overlays such as the body position icon) contain no diagnostic information. During conversion, these pixels were zeroed out so that nnU-Net’s automatic nonzero-mask normalization excludes non-scan-sector regions. Outline masks were also saved separately for potential use in future augmentation experiments.

Segmentation labels were rendered from the expert-annotated polygons as pixel masks with three classes: background (0), liver (1), and mass (2). Mass polygons were drawn over liver polygons so that mass pixels overwrite liver pixels where they overlap.

Apart from the outline masking, no modifications were made to nnU-Net’s default augmentation pipeline, which includes rotations, scaling, elastic deformations, gamma corrections, Gaussian noise, mirroring, and low-resolution simulation.

3.3 nnU-Net configuration

We used nnU-Net v2 in its default 2D configuration. The framework automatically determined a patch size of 512 x 768, batch size of 8, and an 8-stage PlainConvUNet encoder with feature counts of 32, 64, 128, 256, 512, 512, 512, 512. ZScore normalization was applied using the nonzero mask. The default augmentation pipeline was used without modification, as described above.

A custom trainer limited training to 150 epochs, a choice we justify in section 6 (the nnU-Net default is 1000). Checkpoint selection used the exponential moving average (EMA) of pseudo Dice computed across both foreground classes (liver and mass), averaged together. This means the selected “best” checkpoint optimizes for the combined liver-mass performance rather than mass detection specifically, a limitation discussed in Section 7.

3.4 Nested subset design

To isolate the effect of training set size, we created nested stratified subsets at sizes of 25, 50, 100, 200, 300, 400, and 500 images. Each smaller subset is fully contained within every larger subset: the 25-image set is a subset of the 50-image set, which is a subset of the 100-image set, and so on. This ensures that performance differences between sizes reflect additional data, not different data. Stratification preserved the ratio of mass-present to mass-absent cases at each size. All subsets were drawn from the 625-image training set using a fixed random seed (42).

3.5 Evaluation metrics

We report the Dice similarity coefficient for liver segmentation (treating both liver and mass as foreground) and mass segmentation (class 2 only). Liver Dice was computed over all 110 test images. Mass Dice was computed only over the 95 test images containing annotated masses (65 malignant, 30 benign).

For detection rate analysis, a mass was counted as detected if the model predicted any mass (class 2) pixels for that case. False positive analysis on normal cases used the same criterion.

For the size-class interaction analysis, test cases with masses were divided into quartiles by ground truth mass pixel count.

4 Data efficiency results

4.1 Liver segmentation across training sizes

Liver segmentation is data-efficient and effectively solved at 200 training images.

Size	Liver Dice
25	0.820
50	0.846
100	0.878
200	0.893
300	0.898
400	0.889
500	0.899
625	0.899

From 200 to 625 images, liver Dice improves by only 0.006 (0.893 to 0.899). The liver boundary is a high-contrast, consistent anatomical structure on B-mode ultrasound, and the model learns to delineate it reliably with limited training data.

Liver segmentation performance is consistent across pathology types. At 625 training images, normal livers achieve the highest Dice (0.930, std 0.027), followed by malignant (0.901, std 0.091) and benign (0.882, std 0.108).

Liver detection rate is 100%: every test image received some liver prediction.

4.2 Mass segmentation across training sizes

Mass segmentation is substantially harder and remains data-hungry through the largest training set.

Size	Mass Dice
25	0.292
50	0.358
100	0.506
200	0.546
300	0.559
400	0.587
500	0.639
625	0.652

Mass Dice improves at every increment from 25 to 625 images. Unlike liver segmentation, there is no plateau: the curve is still climbing at 625 images, suggesting that additional training data would continue to improve performance. The jump from 25 to 100 images is the steepest (0.292 to 0.506), after which gains become more gradual.

However, these aggregate numbers mask a difference between malignant and benign mass detection, explored in the next section.

5 Failure analysis

5.1 Mass detection by pathology type

Breaking down mass Dice by pathology type reveals two fundamentally different problems.

Size	Benign Dice (std)	n	Malignant Dice (std)	n
25	0.169 (0.236)	30	0.349 (0.234)	65
50	0.147 (0.231)	30	0.455 (0.301)	65
100	0.264 (0.324)	30	0.618 (0.310)	65
200	0.311 (0.327)	30	0.654 (0.282)	65
300	0.317 (0.350)	30	0.671 (0.288)	65
400	0.341 (0.377)	30	0.701 (0.297)	65
500	0.319 (0.337)	30	0.787 (0.213)	65
625	0.403 (0.344)	30	0.767 (0.221)	65

Size	Benign Dice (std)	n	Malignant Dice (std)	n
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Malignant mass Dice improves steadily and reaches 0.77 at 625 images. Benign mass Dice remains below 0.41 across all sizes despite tripling the training data from 200 to 625 images.

The difference in mass Dice between malignant and benign cases is statistically significant (Mann-Whitney U, $p < 0.001$).

At the per-case level, the 625-image model detects 97% of malignant masses (63/65) but only 70% of benign masses (21/30). Among detected malignant cases, mean Dice is 0.79. Among detected benign cases, mean Dice is 0.58.

Category	Detected	Total	Rate
Malignant	63	65	97%
Benign	21	30	70%
Normal (false positives)	3	15	20%

Per-image analysis reveals a bimodal pattern in benign detection: the model either segments a benign mass well (Dice > 0.85) or misses it entirely (Dice = 0.0). Of 30 benign test cases, 9 received a Dice of exactly 0.0, meaning the model predicted no mass pixels at all.

Isoechoic presentation was hypothesized to explain poor benign Dice (Schima et al., 2018), but intensity contrast between mass and surrounding liver parenchyma did not correlate with segmentation performance ($r=-0.087$, $p=0.40$). Size appears to be the dominant factor.

5.2 Mass detection by lesion size

Mass Dice correlates significantly with lesion size (Pearson $r = 0.349$, $p = 0.0005$).

Benign masses occupy a median 5.6% of the visible liver area compared to 15.4% for malignant masses. This matters because mass segmentation performance drops substantially below approximately 5% of liver area (mean Dice 0.44 vs 0.71), a threshold that captures most benign masses but few malignant ones.

The size disparity between benign and malignant masses in AUL is consistent with clinical presentations: common benign liver tumors are typically small and asymptomatic (Oldhafer et al., 2020), while malignant masses tend to be larger at detection.

Quartile	Mean Dice	Std	n	Benign	Malignant
Q1 (smallest)	0.394	0.317	24	17	7
Q2	0.572	0.315	24	8	16
Q3	0.829	0.219	23	2	21

Quartile	Mean Dice	Std	n	Benign	Malignant
Q4 (largest)	0.821	0.128	24	3	21

The smallest quartile (Q1) has a mean Dice of 0.394 compared to 0.829 in Q3. However, size and pathology type are confounded in this dataset: the smallest quartile is 71% benign (17/24), while Q3 and Q4 are over 90% malignant.

Benign masses are proportionally smaller relative to their liver (median mass/liver ratio 0.06 for benign vs 0.15 for malignant), confirming that the size disadvantage is not an artifact of image scale.

5.3 Size-class interaction

To disentangle size from pathology type, we compared benign and malignant Dice within the same size quartiles.

Quartile	Benign Dice	n	Malignant Dice	n
Q1 (smallest)	0.311	17	0.596	7
Q2	0.314	8	0.701	16
Q3	0.937	2	0.819	21
Q4 (largest)	0.809	3	0.823	21

At small sizes (Q1 and Q2), malignant masses substantially outperform benign: 0.60 vs 0.31 in Q1, 0.70 vs 0.31 in Q2. At large sizes (Q3 and Q4), performance converges. This suggests that pathology type matters independently of lesion size, though the small benign sample sizes in Q3 (n=2) and Q4 (n=3) mean this conclusion rests on very limited evidence at larger lesion sizes.

5.4 False positive analysis on normal livers

The model is highly specific on normal livers. Of 15 normal test cases (no mass present), 12 had zero false mass predictions. The remaining 3 had negligible false positive pixel counts: 469 pixels (0.11% of image), 239 pixels (0.06%), and 1,392 pixels (0.32%). These likely represent noise at tissue boundaries rather than hallucinated masses.

6 Projected performance

Fitting a logarithmic curve ($\text{Dice} = a * \ln(\text{size}) + b$) to the data efficiency results provides estimates for performance at larger training set sizes.

Size	Liver Dice	Mass Dice	Benign Mass	Malignant Mass
625 (actual)	0.899	0.652	0.403	0.767

Size	Liver Dice	Mass Dice	Benign Mass	Malignant Mass
1,000 (projected)	0.919	0.707	0.407	0.846
2,000 (projected)	0.936	0.783	0.456	0.935
5,000 (projected)	0.957	0.884	0.520	0.990

Fit quality: mass Dice $R^2 = 0.96$, malignant mass $R^2 = 0.95$, benign mass $R^2 = 0.89$, liver Dice $R^2 = 0.89$.

The projections suggest markedly different data requirements by task:

- 0.90 overall mass Dice would theoretically require 5,800 training images
- 0.90 malignant mass Dice would require 1,500 training images
- 0.70 benign mass Dice would theoretically require 63,000 training images

The benign projection is tenuous but serves to underscore that benign mass detection may not be solvable by collecting more data of the same type.

7 Ablation experiments

Several training runs in the efficiency study peaked near the 150-epoch cap (e.g., 100 images at epoch 148, 300 images at epoch 146, 500 images at epoch 147), raising the question of whether additional training would improve results. We also tested whether nnU-Net’s newer residual encoder architecture (ResEnc M) offers an advantage over the default plain convolutional encoder. Both experiments used the 200-image subset.

7.1 Extended training (300 epochs)

Doubling the epoch budget from 150 to 300 did not improve mass detection. Mass Dice decreased slightly from 0.546 to 0.525, suggesting mild overfitting. Liver Dice was unchanged (0.893).

7.2 Residual encoder architecture

Replacing the plain convolutional encoder with nnU-Net’s ResEnc M preset yielded a marginal improvement in mass Dice (0.556 vs 0.546) that is not statistically meaningful. Liver Dice was unchanged (0.890). ResEnc M increased per-epoch training time by 70% (36s vs 21s per epoch) with no corresponding performance gain.

7.3 Summary

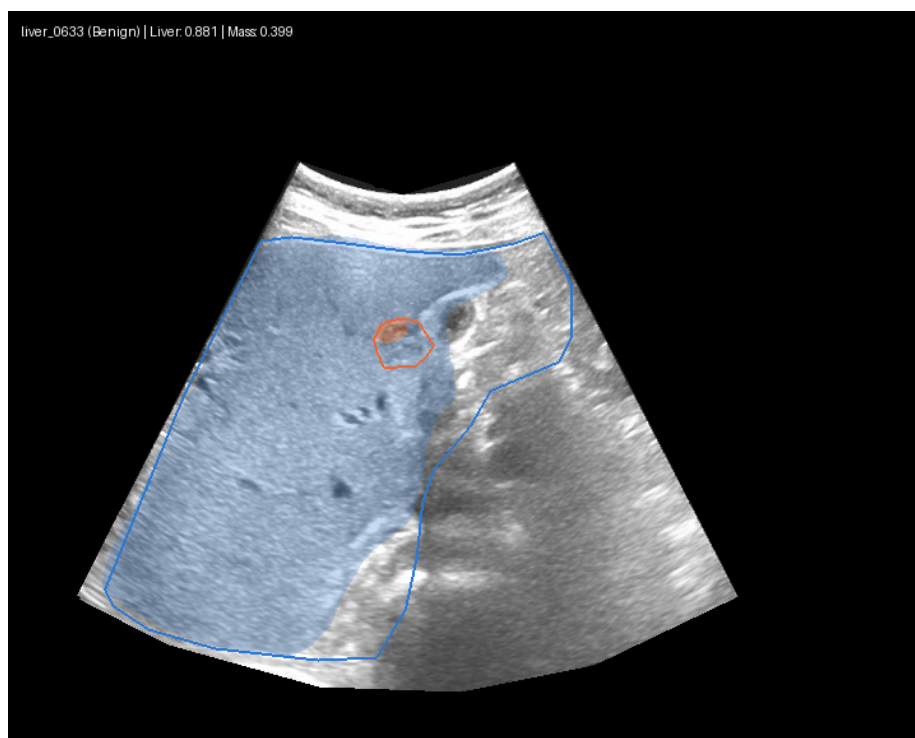


Figure 3: This overlay the 625-image model's predicted mass segmentation sits close to the mean Dice of 0.403, at 0.399. It happens to typify the typically smaller mass sizes of benign masses.

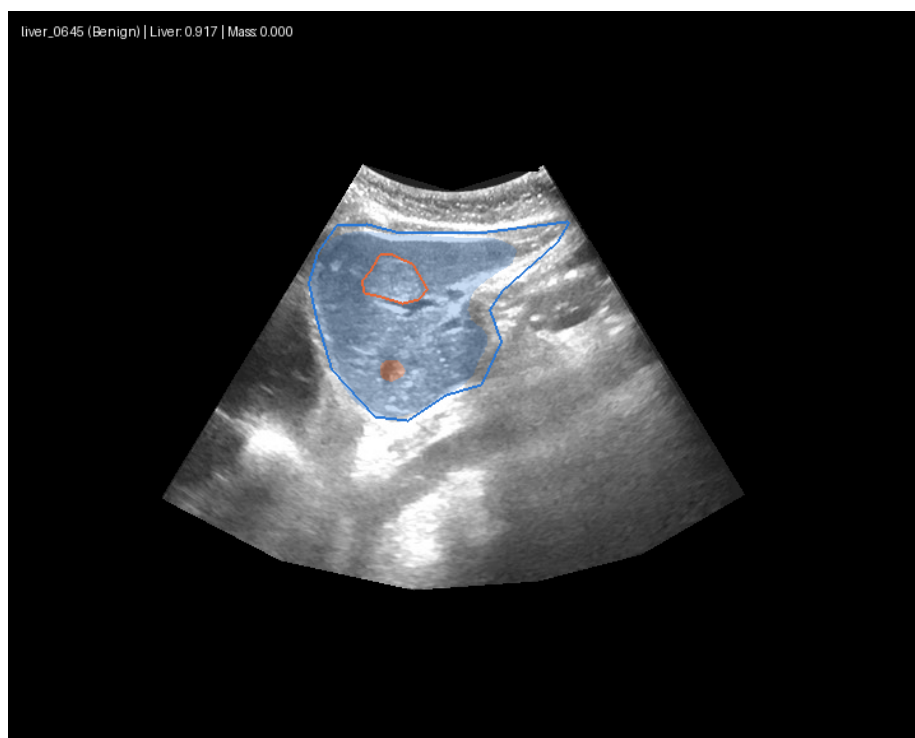


Figure 4: This overlay the 625-image model’s predicted mass segmentation demonstrates that in some cases, particularly with benign masses, the model completely misses the mark.

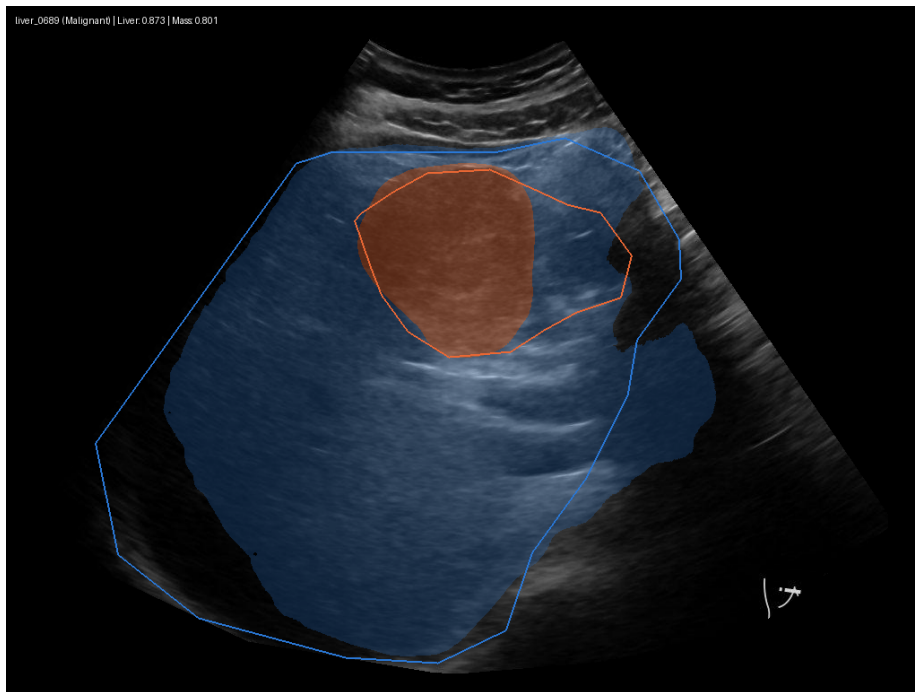


Figure 5: With Dice of 0.873, this overlay the 625-image model's predicted mass segmentation shows that the model not only detects 97% of malignant masses but segments them fairly accurately.

Configuration	Liver Dice	Mass Dice
PlainConvUNet, 150 epochs (baseline)	0.893	0.546
PlainConvUNet, 300 epochs	0.893	0.525
ResEnc M, 300 epochs	0.890	0.556

Neither extended training nor a deeper encoder architecture meaningfully improved mass detection at 200 training images. The most likely bottleneck at this dataset size is the training data itself, as opposed to model capacity or training duration.

8 Discussion

8.1 Clinical implications

If deployed in a screening or monitoring context, the model’s failure profile would be clinically favorable. Malignant mass detection, the most dangerous failure mode, performs well: 97% detection rate with 0.77 Dice at 625 images, and usable performance (0.65 Dice) with as few as 200 training images. The model’s primary weakness, poor benign mass detection, involves the lesion type least likely to change clinical management. Small benign liver lesions are typically incidental findings that do not require intervention.

The model’s putative clinical purpose would be to flag suspicious findings for follow-up imaging in a monitoring or screening capacity, not to provide diagnosis. A malignant mass flagged by the model, even if imprecisely segmented, might trigger the same clinical response: referral for contrast-enhanced CT, providing a more precise characterization.

The cost of a false negative on a benign lesion is a missed incidental finding. The cost of a false negative on a malignant mass is a missed cancer. The model’s asymmetry aligns with this clinical asymmetry.

For false positives on normal livers, the model’s high specificity (80% of normal cases had zero false mass predictions, with negligible pixel counts in the remaining 20%) means unnecessary follow-up imaging would be rare.

8.2 Limitations

This study uses a single dataset (AUL) from a single institution. The data has a Bronze rating according to the broad review of public ultrasound datasets performed by Alsharid et al. Generalization to other ultrasound machines, patient populations, and annotation styles is untested. The 85/15 train/test split uses a single fold rather than full cross-validation.

Checkpoint selection was based on the mean EMA pseudo Dice across liver and mass classes. Because liver Dice plateaus early and dominates the average, the selected checkpoint may not represent peak mass detection performance. Future work should use mass-specific checkpoint selection. Liver and malignant

mass Dice are uncorrelated at the per-image level ($r=-0.113$, $p=0.38$), further supporting the case for independent checkpoint selection by task.

Training was capped at 150 epochs. Several runs peaked near this limit, though ablation experiments at 300 epochs showed no improvement, suggesting the cap did not meaningfully constrain results.

Each training size used a single random seed (42); initialization sensitivity could in theory shift absolute Dice values, as demonstrated by Hänsch et al., but we see consistent trends across nested subsets.

Benign mass analysis is limited by small sample size (30 test cases) and confounded by the correlation between pathology type and lesion size in this dataset. The benign sample sizes in the upper size quartiles ($n=2$ and $n=3$) are too small for robust conclusions about large benign mass detection.

The projected performance estimates in Section 5 are based on log-curve extrapolation from 8 data points, extended well beyond the observed range. They should be treated as rough estimates, not predictions.

8.3 Future work

Several directions could improve on these baselines. Mass-specific checkpoint selection, suggested by the weak correlation between liver and mass Dice, may recover performance lost to the current joint optimization. Augmentation strategies beyond nnU-Net’s defaults, including those explored by [Tupper and Gagné \(2025\)](#), and loss function weighting toward the mass class are natural next steps. The lightweight 2D PlainConvUNet architecture selected by nnU-Net for this dataset is compatible with edge deployment on mobile devices paired with POCUS probes, a direction we plan to explore. Contrast-enhanced or Doppler imaging could improve benign mass detection by providing discriminative features absent in B-mode.

9 Conclusion

This study establishes resource-efficient baselines for liver mass segmentation on B-mode ultrasound using nnU-Net in its default configuration with minimal training duration. The central finding is that malignant and benign masses have fundamentally different data requirements: malignant mass detection reaches clinically usable levels (0.77 Dice, 97% detection rate) with 625 training images, while benign mass detection remains poor (0.40 Dice) and is unlikely to improve through data scaling alone. These baselines provide a reproducible foundation against which future improvements in augmentation, architecture, and imaging modality can be measured.

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